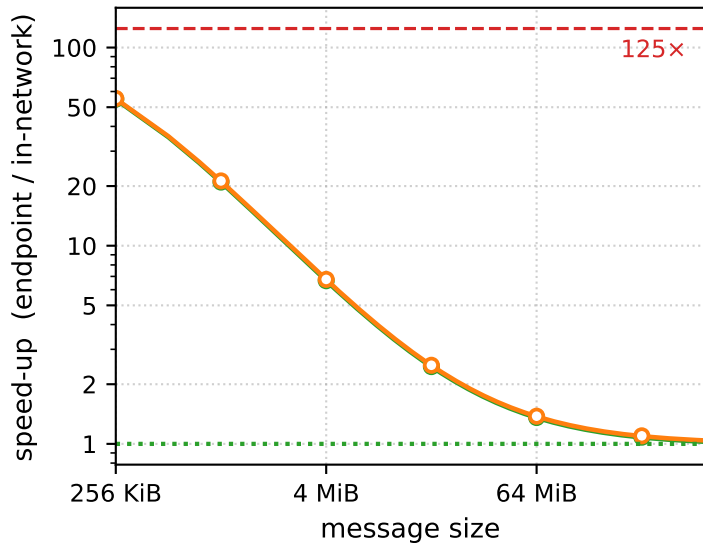
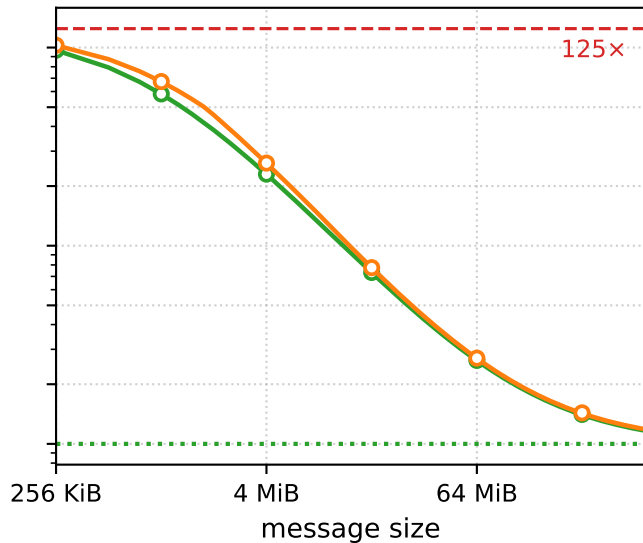


(a) single switch, $d = 1$



(b) three-tier, $d = 3$



— Reduce-Scatter, model
○ Reduce-Scatter, measured

— AllGather, model
○ AllGather, measured

--- latency bound
..... bandwidth reference